Year 2 Statistics and Mechanics

Chapter 6

Projectiles

Practice Booklet

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6.1 Horizontal projection

Practice Book, printed page 49.

1. A particle is projected horizontally at 15 m s^{-1} from a point 50 m above a horizontal surface.

a. Show that the time taken for the particle to reach the surface is $\frac{10\sqrt{5}}{7}$ seconds.

b. Work out the horizontal distance travelled by the particle.

Hint

Draw a diagram that shows all the information given in the question.

2. A particle is projected horizontally from point P at 30 m s^{-1}. Find an expression for:

a. the horizontal distance of the particle from P after t seconds

b. the vertical distance of the particle from P after t seconds.

Hint

Use $s = ut$ and $s = \frac{1}{2}at^2$ in the horizontal and vertical planes respectively to find the horizontal and vertical distances of the particle from P.
← Year 1, Sections 9.3, 9.4

3. A particle is projected horizontally at 40 m s^{-1} from a point P, which is h m above a horizontal plane. The particle hits the plane a horizontal distance of 100 m from P. Work out:

a. the time taken for the particle to land

b. the value of h.

Hint

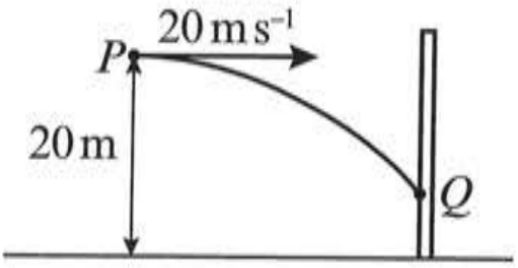
The horizontal motion is modelled as having zero acceleration. Use $s = ut$ in the horizontal direction to find the time taken for the particle to land.

4. E A particle is projected horizontally at u m s^{-1} from a point P, which is 15 m above a horizontal plane. The particle hits the plane a horizontal distance of 24.5 m from P.

a. Work out the value of u. (4 marks)

b. State one modelling assumption used in your answer to part **a**. (1 mark)

5. E/P A point P lies 20 m above a horizontal surface. A ball is thrown horizontally at 20 m s^{-1} from P towards a vertical wall that stands on the horizontal plane. After 2 seconds the ball strikes the wall at the point Q. The ball is modelled as a particle moving freely under gravity.



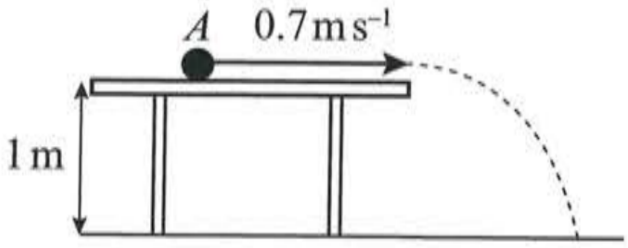
a. Work out the horizontal distance from P to the wall (2 marks)

b. Find the height of Q above the plane. (2 marks)

The model is adjusted to take air resistance into account.

c. Explain briefly how this will affect the height at which the ball strikes the wall. (1 mark)

6. E/P A particle is projected from rest from a point A along a smooth horizontal table at 0.7 m s^{-1}. The table is 1 m high and stands on horizontal ground. Given that it takes 2 seconds from the moment of projection until the ball hits the ground, work out the distance from the point A to the edge of the table. (4 marks)



7. E/P A particle of mass 0.5 kg is projected along a horizontal table with an initial velocity of 6 m s^{-1}. The table has a rough surface. The coefficient of friction between the particle and the table's surface is 0.2.

After travelling for 2 seconds the particle leaves the table.

a. Find the speed of the particle at the instant it leaves the table. (3 marks)

Given that the particle lands a horizontal distance of 0.78 m from the edge of the table,

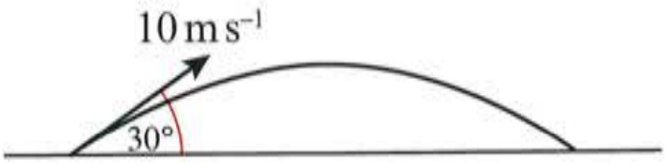
b. work out the height of the table. (3 marks)

c. How does your answer to part **b** change if the horizontal surface of the table is smooth? (1 mark)

6.2 Horizontal and vertical components

Practice Book, printed page 50. In these questions you may assume that **i** and **j** are unit vectors acting horizontally and vertically respectively.

1. A particle is projected from a point on a horizontal plane with an initial velocity of 10 m s^{-1} at an angle of 30° above the horizontal.



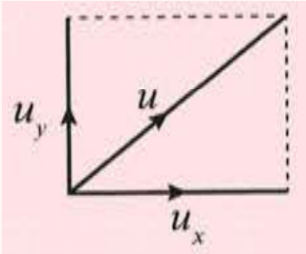
Find:

a. the horizontal component of the initial velocity

b. the vertical component of the initial velocity.

Hint

Resolve horizontally and vertically to find the horizontal and vertical components of the initial velocity.



2. A particle is projected from a point on a horizontal plane with an initial velocity of 20 m s^{-1} at an angle of 45° above the horizontal. Express the initial velocity as a vector in terms of **i** and **j**.

Hint

Find the horizontal and vertical components of the initial velocity, then write them in vector form.

3. A particle is projected with an initial velocity of $(12\mathbf{i} + 5\mathbf{j})$ m s^{-1}. Find the initial speed of the particle and its angle of projection.

Hint

Speed is the magnitude of the velocity vector.
← Year 1, Section 8.4

4. E A particle is projected from a point on a horizontal plane with an initial velocity of 15 m s^{-1} at an angle of 25° above the horizontal.

a. Find the horizontal and vertical components of the initial velocity. (2 marks)

b. Express the initial velocity as a vector in terms of **i** and **j**. (2 marks)

5. E/P A particle is projected from a tall building with an initial velocity of 8 m s^{-1} at an angle of 10° below the horizontal.

a. Find the horizontal and vertical components of the initial velocity. (2 marks)

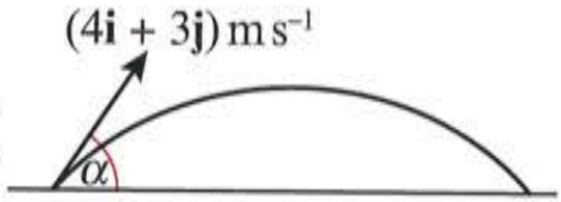
b. Express the initial velocity as a vector in terms of **i** and **j**. (2 marks)

6. E A particle is projected from horizontal ground with an initial velocity of 156 m s^{-1} at an angle α above the horizontal where $\tan\alpha = \frac{12}{5}$

a. Find the horizontal and vertical components of the initial velocity. (2 marks)

b. Express the initial velocity as a vector in terms of **i** and **j**. (2 marks)

7. E A particle is projected with an initial velocity of $(4\mathbf{i} + 3\mathbf{j})$ m s^{-1}.
Find the initial speed of the particle and its angle of projection, α. (3 marks)



6.3 Projection at any angle

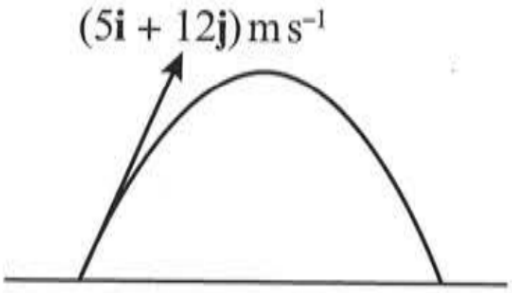
Practice Book, printed page 51. In these questions you may assume that **i** and **j** are unit vectors acting horizontally and vertically respectively.

1. A particle is projected from a point on a horizontal plane with a speed of 98 m s^{-1} at an angle of 30° above the horizontal. Given that the particle moves freely under gravity, find the time of flight of the particle.

Hint

You only need to consider the vertical motion of the particle.

2. A particle is projected from a point on a horizontal plane with a velocity of $(5\mathbf{i} + 12\mathbf{j})$ m s^{-1}. The particle moves freely under gravity.



a. By considering the vertical motion of the particle, find the time of flight of the particle.

b. By considering the horizontal motion of the particle, find the range of the particle.

Hint

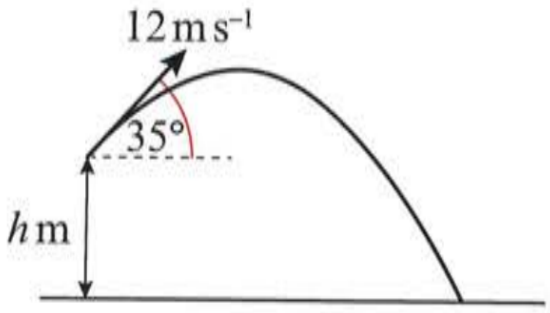
The range is the horizontal distance travelled by the particle.

3. A particle is projected from a point on a horizontal plane with a speed of 34 m s^{-1} at an angle of 50° above the horizontal. Given that the particle moves freely under gravity, find the maximum height reached by the particle.

Hint

The particle reaches its maximum height when the vertical component of its velocity is 0.

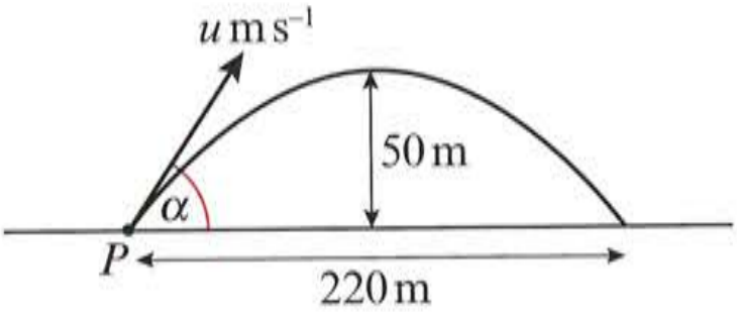
4. E A particle is projected from a point h m above a horizontal plane with a speed of 12 m s^{-1} at an angle of 35° above the horizontal. The particle moves freely under gravity and lands on the horizontal plane after 5 seconds. Find the value of h. (3 marks)



5. E A particle is projected from a point 10 m above a horizontal plane with a speed of 50 m s^{-1} at an angle of 20° above the horizontal. Given that the particle moves freely under gravity, find the range of the particle. (6 marks)

6. E A particle is projected from a point on a horizontal plane with a velocity of $(20\mathbf{i} + 15\mathbf{j})$ m s^{-1}. Given that the particle moves freely under gravity, find the length of time for which the particle is above 10 m. (6 marks)

7. E/P A particle is projected from a point P on horizontal ground with a speed of u m s^{-1} at an angle α above the horizontal. The maximum height reached by the particle is 50 m and the range of the particle is 220 m.



Find the value of:

a.

i. α

ii. u

(10 marks)

b. Work out the speed of the particle after 5 seconds. (4 marks)

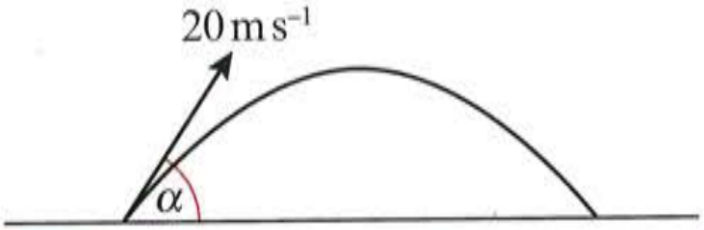
8. E/P The point P lies 200 m from the base of a vertical cliff of height 100 m. An object is projected from point P, towards the cliff, with a velocity of $(36\mathbf{i} + 48\mathbf{j})$ m s^{-1}.
Find the distance the object lands from the edge of the cliff. (6 marks)

9. E/P A rugby player is attempting to kick a penalty. In order for the penalty to be successful, the ball must pass between the posts and clear a crossbar that is 4 m high. The player kicks the ball from the ground with a speed of 25 m s^{-1} at an angle of 40° above the horizontal. Assuming that the ball passes between the posts and by modelling the rugby ball as a projectile moving freely under gravity, work out the greatest distance from which the player can kick a successful penalty. (6 marks)

6.4 Projectile motion formulae

Practice Book, printed page 52. In these questions you may assume that **i** and **j** are unit vectors acting horizontally and vertically respectively.

1. P A particle is projected from a point on a horizontal plane with a speed of 20 m s^{-1} at an angle α above the horizontal.
Given that the particle moves freely under gravity, show that the greatest height it reaches is $\dfrac{200\sin^2\alpha}{g}$



Hint

Consider the vertical motion of the particle and use $v^2 = u^2 + 2as$.

2. P A particle is projected from a point on a horizontal plane with a speed of U at an angle α above the horizontal.

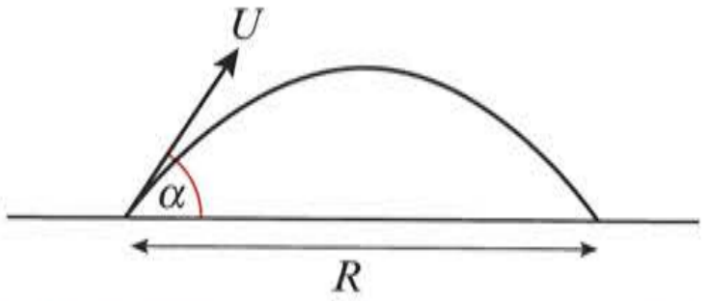
a. Find an expression in terms of U, g and α for the range, R, of the particle.

b. Work out the value of α that gives the greatest possible range.

Hint

Consider the value of α in the range $0 \leq \alpha \leq 90°$ that maximises the value of the trigonometric function in your answer to part **a**.

3. P A particle is projected from a point on a horizontal plane with a speed of U at an angle α above the horizontal. The particle moves freely under gravity, and travels a horizontal distance of R.



a. Show that the maximum height h of the particle is given by $h = R\tan\alpha - \dfrac{gR^2}{8U^2\cos^2\alpha}$

Hint

If the upwards direction is taken to be positive then the vertical acceleration is $-g$.

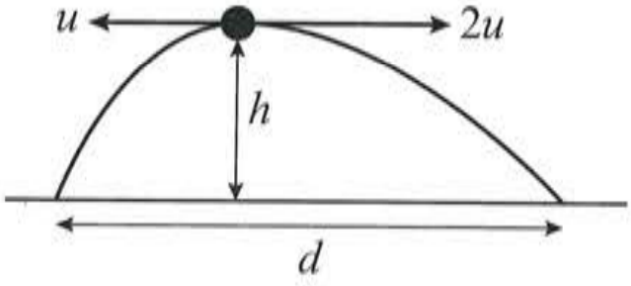
b. Show that the equation found in part **a** can be rewritten as $h = R\tan\alpha - \dfrac{gR^2}{8U^2}(1+\tan^2\alpha)$

Hint

You'll need to use the trigonometric identities covered in the Pure part of your course to answer this.
← Pure Year 2, Chapter 6

4. E/P A particle A is projected from a point P on a horizontal plane with a speed of u_A at an angle α_A above the horizontal. A second particle B is then projected from P at a speed of u_B at an angle α_B above the horizontal. Find, in terms of u_A, u_B, α_A, α_B and g, the distance between the points where the particles land. (4 marks)

5. E/P Two warning flares F_1 and F_2 are fired horizontally in opposite directions from a helicopter that is hovering stationary above level ground at a height of h. F_1 is fired with speed u and F_2 is fired with speed $2u$.
Find, in terms of u, g and h, the distance d between the points where the flares land. (4 marks)



6. E/P A particle is projected from a point P that lies on level ground with velocity $(a\mathbf{i} + b\mathbf{j})$.

a. Show that the horizontal distance travelled by the particle is $\dfrac{2ab}{g}$ (6 marks)

b. Given that the particle travels a horizontal distance of 200 m and that its velocity of projection is parallel to the vector $12\mathbf{i} + 5\mathbf{j}$, find the initial speed of the particle. (4 marks)

7. E/P A ball is thrown from a point P on horizontal ground with a speed of U at an angle α above the horizontal. Given that the ball is modelled as a particle moving freely under gravity, show that the horizontal distance travelled by the ball is $\dfrac{U^2 \sin 2\alpha}{g}$ (6 marks)

8. E/P A particle is projected from a point P 15 m above a horizontal plane with a speed of U at an angle α above the horizontal. The particle moves freely under gravity, and just passes over a wall of height 25 m that is a horizontal distance of 40 m from P.

a. Show that $\dfrac{800g}{U^2}(1 + \tan^2 \alpha) - 40 \tan \alpha + 10 = 0$ (6 marks)

b. Given that $U = 40$ m s^{-1}, work out two possible sizes of angle α. (6 marks)

Problem solving: Set A

In these questions, **i** and **j** are the unit vectors horizontally and vertically respectively.

Bronze

A ball is projected from a point on a horizontal plane with a speed of $2U$ at an angle α above the horizontal. By modelling the ball as a particle moving freely under gravity,

a. show that the time between the instant the ball is projected and the instant it lands on the plane is given by $\frac{4U\sin\alpha}{g}$ (4 marks)

b. Hence show that the range of the particle is given by $\frac{4U^2\sin 2\alpha}{g}$ (4 marks)

Silver

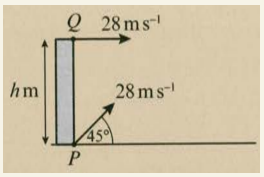
A particle is projected from a point P on a horizontal plane with a speed of U at an angle θ above the horizontal, where $\tan\theta = \frac{3}{4}$. When the particle is a horizontal distance x from P, its height above P is h.

a. Show that $h = ax - b\left(\frac{gx^2}{U^2}\right)$, where a and b are constants to be determined. (6 marks)

b. Given that $U = 35$ m s^{-1}, find the exact speed of the particle at the point where $h = \frac{x}{2}$ (4 marks)

Gold

An arrow is fired horizontally with a speed of 28 m s^{-1} from a point Q at the top of a vertical tower of height h m. Another arrow is simultaneously fired from a point P at the base of the tower with a speed of 28 m s^{-1} at an angle of 45° above the horizontal.
The ground is modelled as a horizontal plane and the two arrows are modelled as particles travelling in the same vertical plane.



a. Given that the paths of the arrows intersect, find the maximum value of h. (4 marks)

A student claims that this means that arrows will collide in mid-air.

b. Explain why the student is incorrect. (1 mark)

c. Find, in terms of h, the horizontal distance from the tower to the point where the paths of the arrows intersect. (6 marks)

Problem solving: Set B

Bronze

A particle is projected from a point 6 m above horizontal ground with a velocity of $(10\mathbf{i} + 15\mathbf{j})$ m s^{-1}. The particle moves freely under gravity. Taking g to be 10 m s^{-2}, work out:

a. the speed of the particle when it hits the ground (4 marks)

b. the length of time for which the particle is travelling at less than 15 m s^{-1}. (6 marks)

Silver

A particle is projected from a point 120 m above horizontal ground with a velocity of $(33\mathbf{i} + 18\mathbf{j})$ m s^{-1}. The particle moves freely under gravity. Taking g to be 10 m s^{-2}, work out

a. the height of the particle when it is moving at a speed of 55 m s^{-1} (5 marks)

b. the angle the velocity of the ball makes with the ground as the ball lands. (4 marks)

Gold

A particle is projected from a point 75 m above horizontal ground with a velocity of $(40\mathbf{i} - 6\mathbf{j})$ m s^{-1}. Taking g to be 10 m s^{-2}, find:

a. the speed of the particle when it hits the ground (4 marks)

b. the distance from the point of projection to the point of landing (4 marks)

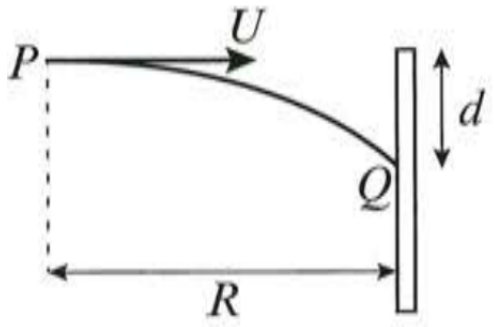
c. the height of the particle at the instant when it is travelling in a direction parallel to the vector $5\mathbf{i} - 4\mathbf{j}$. (4 marks)

d. Show that the particle is never travelling parallel to the vector $\mathbf{i} - \mathbf{j}$. (3 marks)

Selected exam question bank questions

Question 35

A darts player throws a dart at a dartboard which hangs vertically. The motion of a dart is modelled as a particle moving freely under gravity. The dart is projected horizontally with speed U from point P, and moves in a vertical plane which is perpendicular to the plane of the dartboard. It hits the board at a point Q which is a distance d below P.



a. Show that the horizontal distance, R, from the point where the dart was thrown to the dartboard is given by $R = U\sqrt{\frac{2d}{g}}$ (4 marks)

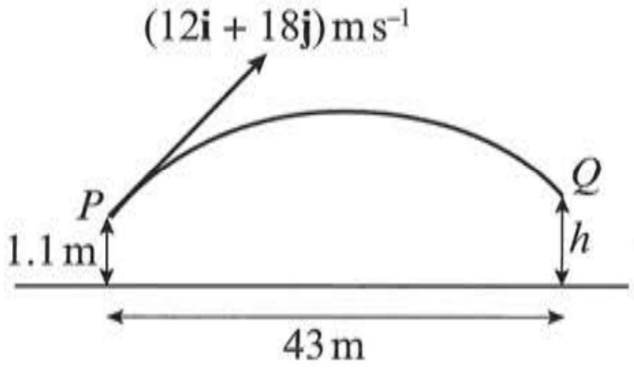
b. Work out the value of d when a dart is thrown horizontally with a speed of 11.2 m s^{-1} from a distance of 2.25 m. (3 marks)

Question 38

A cricketer strikes a cricket ball at a point P which is 1.1 m above horizontal ground.

The unit vectors **i** and **j** are horizontal and vertical respectively.

Immediately after being struck, the ball has velocity $(12\mathbf{i} + 18\mathbf{j})$ m s^{-1}.



The ball moves freely under gravity and can be modelled as a particle.

a. Find the maximum height above the ground reached by the ball. (3 marks)

A fielder stands in the line of flight of the ball at the point Q, a distance of 43 m from P and catches the ball.

b. Work out the height, h, at which the fielder catches the ball. (3 marks)

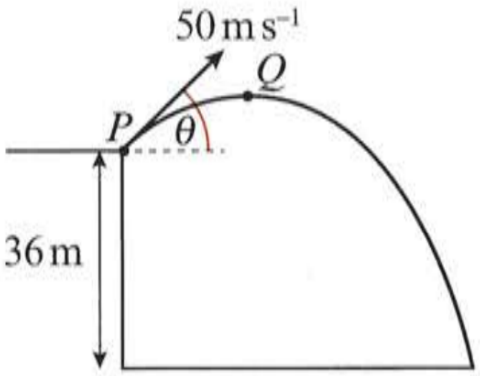
c. State how adjusting the model to take air resistance into account would alter your answer to part b. (1 mark)

Question 42

A particle is projected with speed 50 m s^{-1} from a point P on a cliff that is 36 m above horizontal ground. The particle is projected at an angle θ to the horizontal.

The point Q is the highest point of the path of the ball and is 10 m above the level of P.

The particle moves freely under gravity.



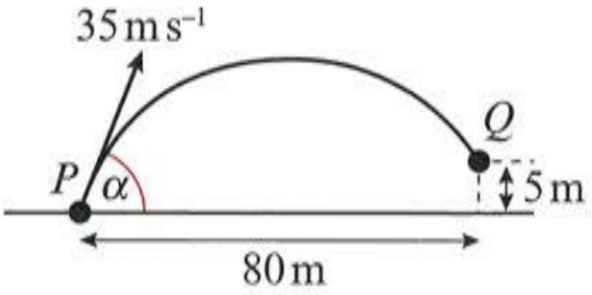
a. Find the size of angle θ. (3 marks)

b. Work out the time after projection at which the particle is moving with its original speed. (4 marks)

c. Calculate the angle that the velocity of the particle makes with ground at the instant it lands. (3 marks)

Question 45

A golfer strikes a golf ball from a point P which lies on horizontal ground. The golfer wants the ball to land close to the hole, which lies a horizontal distance of 80 m from P and a vertical height of 5 m above P. The ball is projected with speed 35 m s^{-1} at an angle α to the horizontal and can be modelled as a particle moving freely under gravity.

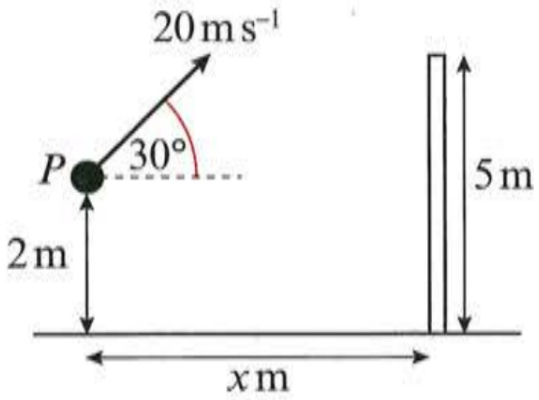


a. Use the model to find the larger angle α that will ensure the golf ball lands on the hole. (8 marks)

b. Comment on the likelihood of the golf ball actually landing on the hole. (1 mark)

Question 48

A ball is thrown with speed 20 m s^{-1} from a point, P, that lies 2 m above horizontal ground. The angle of projection is 30° above the horizontal and the ball moves freely under gravity. A wall of height 5 m lies between P and the point where the ball lands. The horizontal distance from point P to the wall is x m.

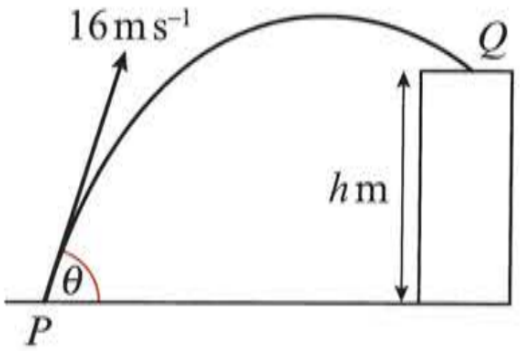


a. Given that the ball clears the wall, find the range of possible values of x. (6 marks)

b. Find the speed of the ball as it hits the ground. (4 marks)

Question 51

A ball is projected with speed 16 m s^{-1} from a point P on horizontal ground. The angle of projection is θ above the horizontal, where $\sin\theta = \frac{4}{5}$. A horizontal platform is at height h metres above the ground. The ball moves freely under gravity until it hits the platform at the point Q. The speed of the ball immediately before it hits the platform at Q is 12 m s^{-1}.



a. Find the value of h. (4 marks)

b. Find the horizontal distance from P to Q. (4 marks)

Answers

6.1 Horizontal projection

1.
- **a.** $u_y = 0$, $a_y = 9.8$ m s^{-2}, $s_y = 50$ m. Substitute into $s = ut + \frac{1}{2}at^2$ to give $50 = \frac{9.8}{2}t^2$. Rearrange to give $t = \frac{10\sqrt{5}}{7}$ seconds.
- **b.** $\frac{150\sqrt{5}}{7}$ m

2.
- **a.** $30t$ m
- **b.** $4.9t^2$ m

3.
- **a.** 2.5 seconds
- **b.** $\frac{245}{8}$

4.
- **a.** 14.0 (3 s.f.)
- **b.** Air resistance negligible.

5.
- **a.** 40 m
- **b.** 0.4 m
- **c.** Q will be closer to the horizontal surface.

6. $\frac{14-\sqrt{10}}{10}$ m

7.
- **a.** 2.08 m s^{-1}
- **b.** $\frac{441}{640}$ m
- **c.** The table height would be less than previously calculated.

6.2 Horizontal and vertical components

1.
- **a.** $5\sqrt{3}$ m s^{-1}
- **b.** 5 m s^{-1}

2. $(10\sqrt{2}\mathbf{i} + 10\sqrt{2}\mathbf{j})$ m s^{-1}

3. 13 m s^{-1} at an angle 22.6° (3 s.f.) above the horizontal

4.
- **a.** 13.6 m s^{-1} (3 s.f.) and 6.34 m s^{-1} (3 s.f.)
- **b.** $(13.6\mathbf{i} + 6.34\mathbf{j})$ m s^{-1}

5.
- **a.** 7.88 m s^{-1} (3 s.f.) and -1.39 m s^{-1} (3 s.f.)
- **b.** $(7.88\mathbf{i} - 1.39\mathbf{j})$ m s^{-1}

6.
- **a.** 60 m s^{-1} and 144 m s^{-1} (3 s.f.)
- **b.** $(60\mathbf{i} + 144\mathbf{j})$ m s^{-1}

7. 5 m s^{-1} at an angle 36.9° (3 s.f.) above the horizontal

6.3 Projection at any angle

1. 10 seconds

2.
- **a.** 2.45 seconds (3 s.f.)
- **b.** 12.2 m (3 s.f.)

3. 34.6 m (3 s.f.)

4. 88.1 (3 s.f.)

5. 188 m (3 s.f.)

6. 1.10 seconds (3 s.f.)

7.
- **a.**
  - **i.** 42.3° (3 s.f.)
  - **ii.** 46.5 (3 s.f.)
- **b.** 38.7 m s^{-1} (3 s.f.)

8. 44.5 m (3 s.f.)

9. 57.6 m (3 s.f.)

6.4 Projectile motion formulae

1. At maximum height $v_y = 0$. Substitute $u_y = 20\sin\alpha$, $a = -g$, $v_y = 0$ into $v^2 = u^2 + 2as$ to give $0 = (20\sin\alpha)^2 - 2gh$. Rearrange to give $h = \frac{200\sin^2\alpha}{g}$.

2.
- **a.** $R = \frac{U^2\sin 2\alpha}{g}$
- **b.** 45°

3.
- **a.** Substitute $u_x = U\cos\alpha$, $a_x = 0$, $s_x = R$ into $s_x = u_x t + \frac{1}{2}a_x t^2$ to give $t = \frac{R}{U\cos\alpha}$.
  This gives the time when the particle lands, so the time when the particle is at maximum height is $\frac{R}{2U\cos\alpha}$.
  Substitute $u_y = U\sin\alpha$, $a_y = -g$, $t = \frac{R}{2U\cos\alpha}$ into $s_y = u_y t + \frac{1}{2}a_y t^2$ to give $h = R\tan\alpha - \frac{gR^2}{8U^2\cos^2\alpha}$.
- **b.** Use $\frac{1}{\cos^2\alpha} = \sec^2\alpha = 1 + \tan^2\alpha$ to give $h = R\tan\alpha - \frac{gR^2}{8U^2}(1 + \tan^2\alpha)$.

4. $\frac{1}{g}|u_A^2\sin 2\alpha_A - u_B^2\sin 2\alpha_B|$

5. $d = 3u\sqrt{\frac{2h}{g}}$

6.
- **a.** Let d be the range of the projectile and t_1 be the time at which it lands.
  From horizontal motion: $t_1 = \frac{d}{a}$ (1)
  For the vertical motion at time of greatest height:

$$u = b, \quad v = 0, \quad t = \frac{t_1}{2}, \quad a = -g$$

  Substitute into $v = u + at$ to give $0 = b - \frac{gt_1}{2}$ (2)
  Substitute (1) into (2) to give $0 = b - \frac{gd}{2a}$
  Rearrange to give $d = \frac{2ab}{g}$
- **b.** 52.5 m s^{-1} (3 s.f.)

7. Use $u_y = U\sin\alpha$, $a = -g$, $s_y = 0$ to find $t = \frac{2U\sin\alpha}{g}$. Then use $u_x = U\cos\alpha$, $a = 0$, $t = \frac{2U\sin\alpha}{g}$ to find $s = \frac{2U^2\sin\alpha\cos\alpha}{g}$.
Finally, using $\sin 2\alpha = 2\sin\alpha\cos\alpha$ gives $s = \frac{U^2\sin 2\alpha}{g}$.

8.

a. When projectile is at top of the wall, from horizontal motion: $t = \frac{40}{U\cos\alpha}$ (1)
For the vertical motion: $10 = Ut\sin\alpha - \frac{g}{2}t^2$ (2)
Substitute (1) into (2) to give

$$10 = U\sin\alpha\left(\frac{40}{U\cos\alpha}\right) - \frac{g}{2}\left(\frac{40}{U\cos\alpha}\right)^2$$

$$\Rightarrow \frac{800g}{U^2}\left(\frac{1}{\cos^2\alpha}\right) - 40\tan\alpha + 10 = 0$$

$$\Rightarrow \frac{800g}{U^2}(1+\tan^2\alpha) - 40\tan\alpha + 10 = 0$$

b. 21.4°, 82.7° (3 s.f.)

Problem solving: Set A

Bronze

a. $\text{R}(\uparrow) : s = ut + \frac{1}{2}at^2$
$0 = 2U\sin\alpha t - \frac{1}{2}gt^2$
$0 = t(2U\sin\alpha - \frac{1}{2}gt)$
$t = 0$ (point of projection) or $2U\sin\alpha = \frac{1}{2}gt$
$\Rightarrow t = \frac{4U\sin\alpha}{g}$

b. $\text{R}(\rightarrow) : s = ut$
$= 2U\cos\alpha\left(\frac{4U\sin\alpha}{g}\right)$
$= \frac{8U^2\sin\alpha\cos\alpha}{g} = \frac{4U^2\sin 2\alpha}{g}$

Silver

a. $\text{R}(\rightarrow) : u_x = U\cos\alpha = \frac{4}{5}U$
$\text{R}(\uparrow) : u_y = U\sin\alpha = \frac{3}{5}U$
$\text{R}(\rightarrow) : s = ut \Rightarrow x = \frac{4}{5}Ut$ (1)
$\text{R}(\uparrow) : s = ut + \frac{1}{2}at^2 \Rightarrow h = \frac{3}{5}Ut - \frac{1}{2}gt^2$ (2)
Rearrange (1) and substitute into (2)

$$h = \frac{3}{5}U\left(\frac{5x}{4U}\right) - \frac{1}{2}g\left(\frac{5x}{4U}\right)^2$$

$$h = \frac{3}{4}x - \frac{25}{32}\left(\frac{gx^2}{U^2}\right)$$

b. $7\sqrt{17}$ m s^{-1}

Gold

a. 40

b. The arrows will reach the point of intersection at different times.

c. $(80 - 4\sqrt{400 - 10h})$ m

Problem solving: Set B

Bronze

a. 21.1 m s^{-1}

b. 2.24 seconds (3 s.f.)

Silver

a. 39.4 m above ground

b. 57.7° (3 s.f.)

Gold

a. 56 m s^{-1}

b. 152 m (3 s.f.)

c. 25.6 m above ground

d. Using $v^2 = u^2 + 2as$
$(-40)^2 = (-6)^2 - 20s \Rightarrow s = 78.2 > 75$ so the particle hits the ground before achieving a velocity of $(40\mathbf{i} - 40\mathbf{j})$ m s^{-1}.

Selected exam question bank answers

Question 35

a. Use $s = ut + \frac{1}{2}at^2$ horizontally to find $R = Ut$. Use $s = \frac{1}{2}at^2$ vertically to find $t = \sqrt{\frac{2d}{g}}$. Substitute to find $R = U\sqrt{\frac{2d}{g}}$.

b. 19.8 (3 s.f.)

Question 38

a. 17.6 m (3 s.f.)

b. 2.68 m (3 s.f.)

c. Fielder catches the ball nearer the ground, or the ball might not reach the fielder before hitting the ground.

Question 42

a. 16.3°

b. 2.86 seconds (3 s.f.)

c. 32.0° (3 s.f.)

Question 45

a. 69.5°

b. The ball is unlikely to land on the hole exactly. Valid reasons include the model does not consider air resistance and/or rotational motion of the ball, the golfer is unlikely to hit the ball at the exact angle or speed, and the measurements may not be exact.

Question 48

a. $6.3 < x < 29.0$

b. 21.0 m s^{-1} (3 s.f.)

Question 51

a. 5.71 (3 s.f.)

b. 19.6 m (3 s.f.)